

PRACTICAL NO :17

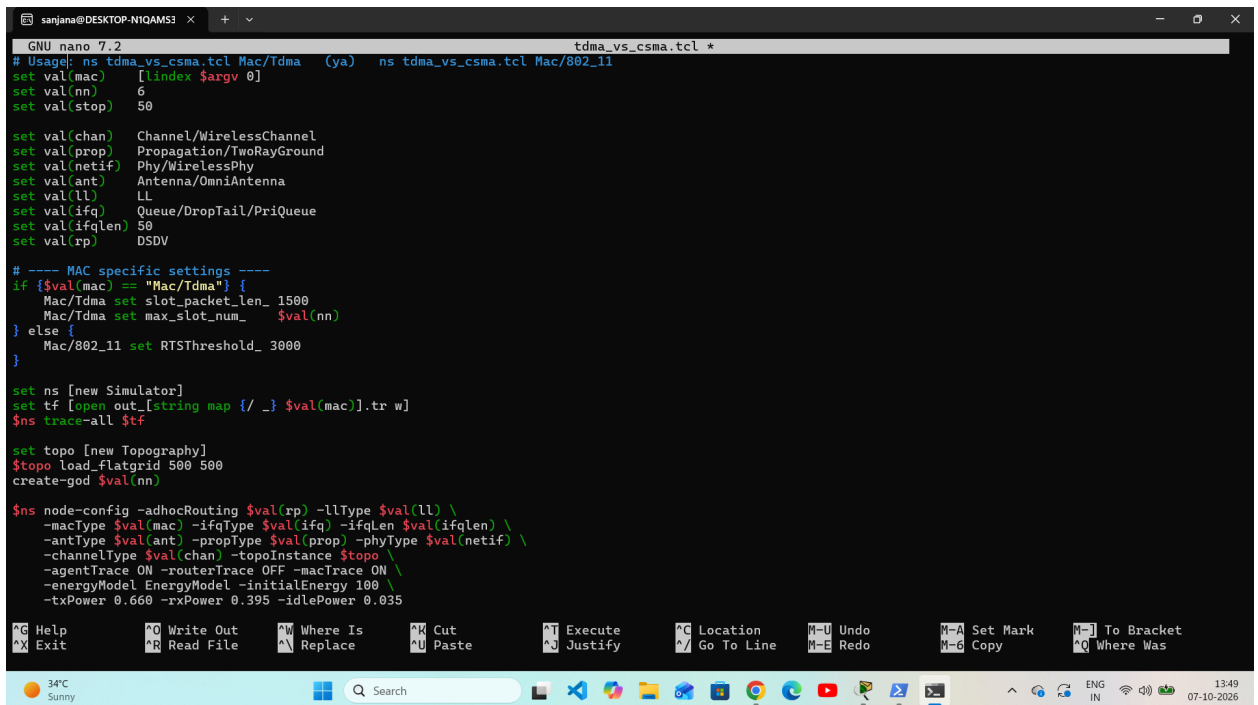
AIM:Simulation of TDMA-based MAC Protocol

Simulate TDMA slot allocation and compare energy efficiency with CSMA.

Created the folder `ns2_lab` and wrote the script with `nano`.

Ran `ns tdma_vs_csma.tcl Mac/Tdma` and `ns tdma_vs_csma.tcl Mac/802_11`.

```
sanjana@DESKTOP-N1QAMS3:~$ mkdir ~/ns2_lab
sanjana@DESKTOP-N1QAMS3:~$ cd ~/ns2_lab
sanjana@DESKTOP-N1QAMS3:~/ns2_lab$ nano tdma_vs_csma.tcl
sanjana@DESKTOP-N1QAMS3:~/ns2_lab$ ns tdma_vs_csma.tcl Mac/Tdma
num_nodes is set 6
warning: Please use -channel as shown in tcl/ex/wireless-mitf.tcl
INITIALIZE THE LIST xListHead
channel.cc:sendUp - Calc highestAntennaZ_ and distCST_
highestAntennaZ_ = 1.5, distCST_ = 550.0
SORTING LISTS ...DONE!
Node 0 residual energy = 97.631352 J
Node 1 residual energy = 99.063168 J
Node 2 residual energy = 99.064428 J
Node 3 residual energy = 99.064275 J
Node 4 residual energy = 99.065459 J
Node 5 residual energy = 99.065535 J
MAC = Mac/Tdma | Total energy consumed = 7.0457829999999859 J
```



```
sanjana@DESKTOP-N1QAMS3:~$ ns tdma_vs_csma.tcl Mac/Tdma
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warning: Please use -channel as shown in tcl/ex/wireless-mitf.tcl
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Noted the residual energy of every node and the total energy consumed.

Counted received packets from the trace files using `grep "^r" out_Mac-xxx.tr | grep AGT | grep -c cbr`

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```
sanjana@DESKTOP-N1QAMS3:~/ns2_lab$ ns tdma_vs_csma.tcl Mac/802_11
num_nodes is set 6
warning: Please use -channel as shown in tcl/ex/wireless-mitf.tcl
INITIALIZE THE LIST xListHead
channel.cc:sendUp - Calc highestAntennaZ_ and distCST_
highestAntennaZ_ = 1.5, distCST_ = 550.0
SORTING LISTS ...DONE!
Node 0 residual energy = 94.462728 J
Node 1 residual energy = 94.094348 J
Node 2 residual energy = 94.079124 J
Node 3 residual energy = 94.056652 J
Node 4 residual energy = 94.060418 J
Node 5 residual energy = 94.042834 J
MAC = Mac/802_11 | Total energy consumed = 35.2038960000000015 J
sanjana@DESKTOP-N1QAMS3:~/ns2_lab$ grep "^r" out_Mac_Tdma.tr | grep AGT | grep -c cbr
1994
sanjana@DESKTOP-N1QAMS3:~/ns2_lab$ grep "^r" out_Mac_802_11.tr | grep AGT | grep -c cbr
2000
```

```
sanjana@DESKTOP-N1QAMS3:~/ns2_lab$ grep "^r" out_Mac_Tdma.tr | grep AGT | grep -c cbr
1994
sanjana@DESKTOP-N1QAMS3:~/ns2_lab$ grep "^r" out_Mac_802_11.tr | grep AGT | grep -c cbr
2000
sanjana@DESKTOP-N1QAMS3:~/ns2_lab$
```

